Induction Heads and Model Collapse: How the Real-to-Synthetic Data Ratio Shapes In-Context Learning Circuits

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Abstract

Model collapse has so far been measured through outputs, which may understate it if internal mechanisms degrade first. This study therefore traces the induction circuit behind in-context learning (ICL) in two-layer transformers retrained on data that the previous model partly or wholly relabels, or that accumulates alongside the real data. A tuned model reproduces its own labels almost exactly, so that accuracy holds under every real-to-synthetic ratio. Once errors are forced through a raised sampling temperature, however, mechanistic changes precede accuracy loss: the logit margin halves and the circuit spreads while accuracy is almost unchanged, and the margin predicts the next generation's error rate, so that errors compound. Retaining real data limits how far accuracy falls, and accumulation delays its decline, but neither stops the margin's erosion. Overall, retaining real data protects a model's behaviour for longer than its circuit, so that collapse may be detected in the circuit before accuracy falls.

Keywords: model collapse; induction heads; mechanistic interpretability

Introduction

Recursively training large language models (LLMs) on data generated by other models causes *model collapse*, a loss of information about the true data distribution (Shumailov et al., 2024), which retaining real data appears to avoid (Gerstgrasser et al., 2024). Both findings, however, rest on model outputs, which would understate collapse if internal mechanisms degrade first.

In-context learning (ICL) lets models adapt to new tasks from prompt examples without parameter updates (Olsson et al., 2022; Singh et al., 2024), and in the task of Singh et al. (2024) it is carried out by an induction circuit whose components are known, so that damage from recursive training can be traced to specific heads rather than inferred from accuracy. This study therefore asks how the share of real data in each generation's training pool shapes this circuit.

Background

Induction heads A previous-token head copies each symbol into the label that follows it, and an induction head matches the query against these copies through its key input and copies the matching label to the output; several such heads can contribute additively (Singh et al., 2024), and their formation tracks ICL ability (Chen et al., 2026).

Collapse and accumulation Collapse arises mainly from sampling error, which loses the tails of the distribution first and compounds across generations (Shumailov et al., 2024). Replacing real data makes test error grow with every iteration, whereas accumulating it keeps the error bounded (Gerstgrasser et al., 2024), a contrast this study tests in the induction circuit.

Methods

This study uses the setup of Singh et al. (2024): each sequence holds two exemplar–label pairs and a query, with Omniglot exemplars (Lake et al., 2015) encoded by a frozen ResNet18 (He et al., 2016) and five labels reassigned per sequence, so that only the context holds the answer. A causal two-layer attention-only transformer with eight heads per layer trains on 50 classes for one million sequences, at learning rates of 10^{-4} (base) and 3×10^{-4} (extended) tuned on 1,473 validation classes, and is tested on 100 unseen classes. The base model predicts the query's label, while the extended model also predicts a next symbol, chosen from the context, and its label, so that three tokens per sequence become synthetic.

Real-to-synthetic ratio Each run trains on a pool of $N = 20{,}000$ sequences for up to 20 generations, the first real, with training length fixed and three seeds per condition. The previous model then regenerates the targets of 0%, 50%, 90%, or 100% of the pool from their real prompts, where 90% matches the 10% of real data kept by Shumailov et al. (2024). Accumulation instead adds N synthetic sequences per generation (Gerstgrasser et al., 2024), matching 50% replacement at generation 1, so that keeping earlier data can be separated from the real share. As the tuned model makes almost no errors when sampling targets at $T = 1$, errors are also forced at $T = 3$ and 5.

Circuit analysis Each generation is analysed on the same real sequences through its margin, the logit lead of the correct label over the other label in the context, and the edges that Automatic Circuit DisCovery (ACDC) (Conmy et al., 2023) keeps at $\tau = 0.001$, which measure how strongly and how sparsely the circuit separates the answer. The induction head's attention to the matching label measures whether it finds the answer, and its direct effect, the margin its output alone contributes, how strongly it copies it.

Results

Table 1: Generation 19 of every 20-generation run, mean ± SD over three seeds, under full replacement unless stated. †Margin below 1, where the circuit size is less reliable.

		Base model			Extended	
Condition	Real	Acc.	Margin	Edges	Margin	Edges
No forced errors ($T = 1$)						
Real only	100%	0.98 ±0.01	17.5 ±2.2	23 ±5	15.7 ±0.3	42 ±7
100% replaced	0%	0.98 ±0.00	12.5 ±0.3	35 ±7	12.8 ±0.9	92 ±36
Accumulate	5%	0.99 ±0.00	9.8 ±1.0	20 ±3	6.3 ±0.1	34 ±5
Forced errors ($T = 3$)						
50% replaced	50%	0.79 ±0.01	2.6 ±0.1	194 ±11	3.8 ±0.3	89 ±19
90% replaced	10%	0.33 ±0.01	0.4 ±0.1	207 ±3†	0.1 ±0.0	180 ±4†
100% replaced	0%	0.19 ±0.02	0.1 ±0.0	207 ±7†	−0.1 ±0.1	163 ±7†
Accumulate	5%	0.92 ±0.02	0.7 ±0.0	17 ±5†	0.6 ±0.0	11 ±2†

No errors, no collapse At generation 0, both models solve the task by ICL, with mean test accuracies of 0.98 and 0.96, perplexities of 1.14 and 1.28, and validation accuracies of 0.99 and

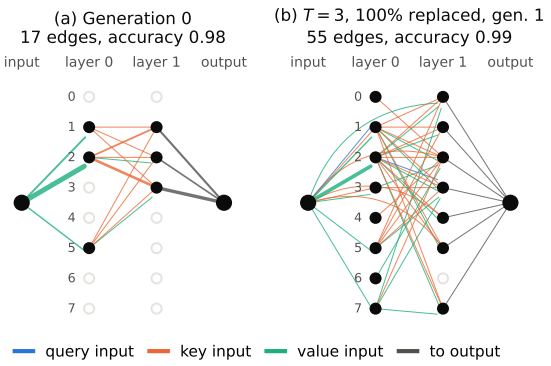


Figure 1: ACDC circuits of the base model (one seed), with filled nodes for heads in the circuit and wider edges for larger effects: the previous-token head A0.2 feeds the induction head A1.3 (a), and one generation of full replacement at $T = 3$ triples the circuit (b).

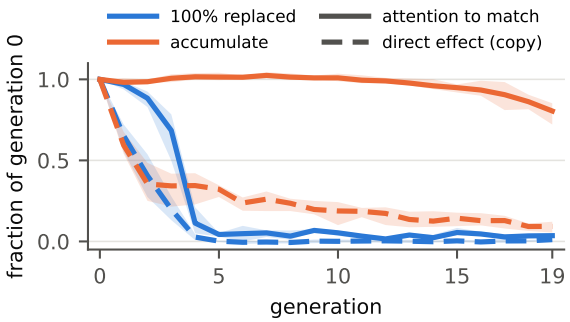


Figure 2: The induction head's attention to the match and direct effect at $T = 3$, relative to generation 0 (three-seed mean and range).

0.97 for the base and extended models. At $T = 1$, only 0.01% of labels are wrong after 20 generations, so that every replacement level yields the same model and accuracy stays at 0.98, although the margin still falls under full replacement (Table 1).

Confidence and circuit change before accuracy With errors forced at $T = 3$, full replacement halves the margin by generation 1, from 18.4 to 10.1, and more than triples the circuit, from 22 to 74 edges on average (17 and 55 in Figure 1), while accuracy stays at 0.95 or above. By generation 3, the circuit spans 211 edges as accuracy falls to 0.66, and the extended model follows the same order.

The induction head: the match holds, the copy weakens Under full replacement (Figure 2), the induction head's attention to the matching label keeps 88% of its generation 0 value by generation 2, whereas its direct effect keeps only 41%, and its attention collapses only at generation 4, after accuracy. Under accumulation, attention to the match stays at 0.89 or above for 15 generations, whereas the direct effect falls by 86%, so that real data preserve where the head looks but not how strongly it copies.

The margin sets the next generation's errors Each model labels the next generation's data, and if its correct label leads the four wrong labels by its margin D_g, sampling at T makes a share

$$\hat{\varepsilon}_{g+1} = \frac{4\,e^{-D_g/T}}{1+4\,e^{-D_g/T}} \tag{1}$$

of these labels wrong, small while D_g is several times T but rising steeply as it shrinks. This predicts the observed share across both models, temperatures, and conditions to within 2.1 percentage points on average ($r = 0.994$), so that a falling margin writes more errors into the next pool and so lowers the next margin, a loop that only real data can slow.

The real share sets how far collapse goes At $T = 3$, each replacement level settles at an accuracy set by its real share, 0.79 with half of the real data, 0.33 with a tenth, and 0.19 with none, with both models' margins in the same order (Table 1), although even half leaves the circuit spread across 194 edges. Accumulation holds accuracy at 0.96 or above for 15 generations, yet its margin falls by 96%, and accuracy declines once the margin drops below 1 by generation 16. The extended model's accuracy settles at 0.84, 0.34, and 0.20, and 0.93 under accumulation. It does not favour some symbols, as its next-symbol entropy stays at 3.86 or above, against 3.91 for a uniform choice, but copies wrongly: under full replacement, 96% of its next symbols fall outside the context and its next-label accuracy falls from 0.96 to 0.20.

Discussion

The real-to-synthetic ratio shaped the induction circuit mainly once the synthetic data carried errors, and the margin then became the route through which collapse compounds, which extends Shumailov et al. (2024) with the internal quantity that sets the sampling error. Within the induction head, the copy failed before the match, which suggests that the copying operation of Singh et al. (2024) is the more fragile. Accumulation protected accuracy, likely because every real prompt keeps its correct label, but not the margin, so that, unlike in Gerstgrasser et al. (2024), it delayed collapse rather than preventing it. The falling margin shows declining confidence, and the spreading circuit and weakening copy a changed computation, both before accuracy fell, so that either could flag collapse early. These conclusions are bounded by forcing errors through the sampling temperature rather than through a weaker model.

Conclusion

This investigation showed that, once synthetic data carry errors, confidence and the induction circuit change before accuracy, while the share of real data sets only how far it falls, from 0.79 with half to chance with none. Overall, real data protect what a model does for longer than how confidently it does it, so that collapse is better detected in the circuit than in accuracy.

Contribution Statements

We, Lily de Melo, Tawananyasha Kadango, and Gopolang Mohlakola, worked together fairly on this project.

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    - The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
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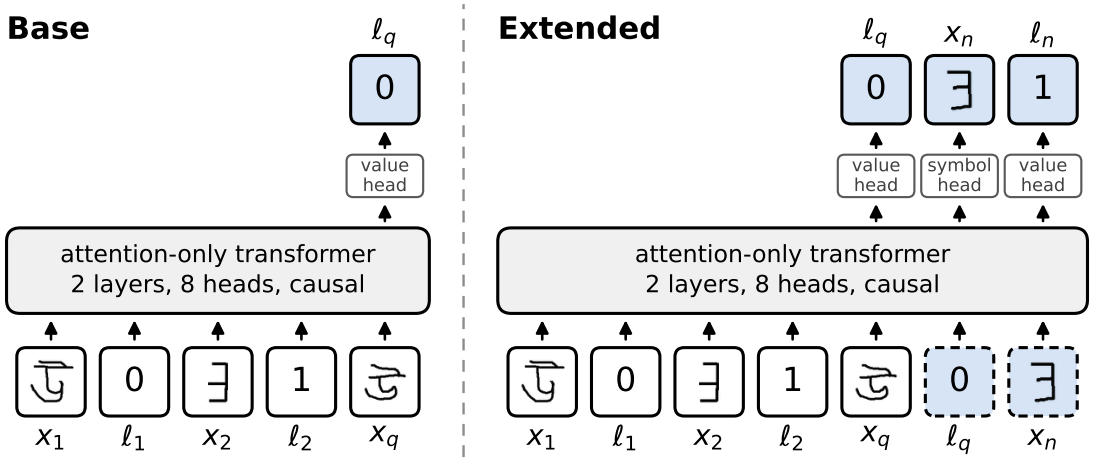


Figure S1: Base (left) and extended (right) models. The base model predicts the query's label from two symbol–label pairs and the query, and the extended model also predicts the next symbol and its label. Shaded boxes are model outputs; dashed inputs are true tokens during training and the model's own samples when it generates the next generation's data.

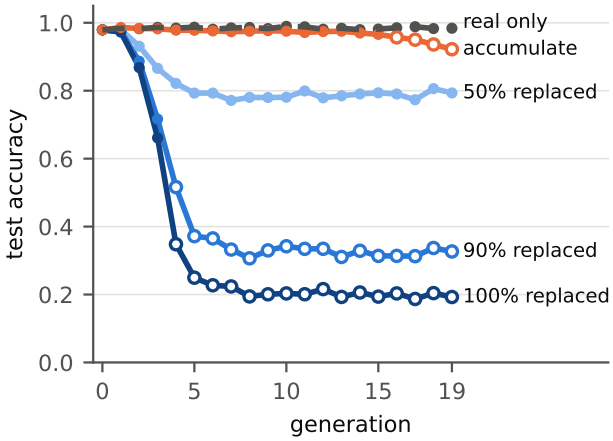


Figure S2: Test accuracy of the base model at $T = 3$, mean of three seeds. Each replacement level settles at an accuracy set by its real share, while accumulation stays with the real-only control for about 15 generations before declining. Hollow markers show a mean margin below 1.

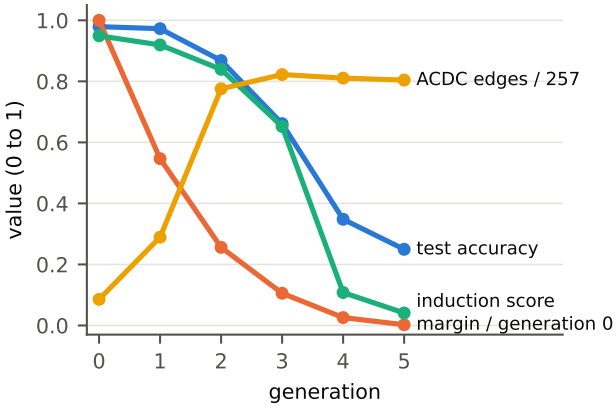


Figure S3: Order of failure of the base model under full replacement at $T = 3$, mean of three seeds. The margin halves and the circuit more than triples by generation 1, whereas accuracy and the induction score fall mainly from generation 3.

Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools [ ]

I **did** make use of generative AI tools for the following (tick all that apply):

1.	Idea Generation (research problem/design, hypothesis)	[ ]
2.	Sourcing Related Work (summarising, identifying sources)	[ ]
3.	Methods and Experiment Design (experiment setup, model tuning)	[X]
4.	Data Analysis (presentation, coding, interpretation)	[X]
5.	Theoretical Development (theorem proving, conceptual analysis)	[ ]
6.	Code Development (generating algorithms, writing scripts)	[X]
7.	Presentation (rendering graphics, formatting)	[X]
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If other uses were involved, please specify below:

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